Cofinal Pell Centers and Oriented Power Certificates

Four explicit Diophantine equations for syntactic ZFC consistency

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An additive contribution to ProveIt

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Abstract

We give two arithmetic reductions and four fully specified integer polynomials whose absence of integer zeros is provably equivalent in ZFC to its syntactic consistency, by composition with the identified ProveIt C222 refutation calibration. First, an even Pell frame permits the free center $A = L^2 + 2u^2$, replacing $L^2 + u^2 + v^2$. The new ingredient is an order-eight identity in a finite quadratic algebra. Canonical parity in the existing source makes the required frame change only a constant increment. The resulting quartics have 229 and 228 integer variables, respectively 4,477 and 12,001 collected monomials, and coefficient height 7,968. Second, oriented reciprocal pooling combines power certificates at different, initially signed bases. Its conclusion applies to nonzero bases while zero bases remain inactive; a parity argument then proves that every actual base is nonzero without circular reasoning. This gives 206 variables at degree 24 and 207 at degree 16. All variables are unrestricted integers. Literal residuals, exact coefficients, reconstruction programs, and independent finite checks accompany the proofs. These are measured improvements over named parents and new tradeoffs; a global simplest equation, worldwide priority, and a new end-to-end proof-assistant derivation are not claimed.

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1 Result, specification, and comparison

1.1 The precise logical target

Fix the effective first-order ZFC presentation called `ZFCax_s` in ProveIt and its finite derivation relation `BProv`. Let $\mathrm{Prf}_{\mathsf{ZFC}}(p, \bot)$ be its arithmetized proof predicate. Throughout,

$$\mathrm{Con}(\mathsf{ZFC}) \;\equiv\; \neg\exists p \in \mathbb{N}\ \mathrm{Prf}_{\mathsf{ZFC}}(p, \bot).$$

Effective translations between ordinary presentations give the usual syntactic consistency statement. No assumption about externally standard integers, transitive set models, or the truth of all ZFC theorems is substituted for this definition.

For each of the four polynomials below the result is

$$\mathsf{ZFC} \vdash (\exists \boldsymbol{x} \in \mathbb{Z}^n\ P(\boldsymbol{x}) = 0) \;\longleftrightarrow\; \exists p \in \mathbb{N}\ \mathrm{Prf}_{\mathsf{ZFC}}(p, \bot). \tag{1}$$

Negating both sides gives the requested no-solution equivalence. The claim is an ordinary mathematical proof formalizable in ZFC, using the explicitly credited inherited calibration; the archive does not contain a newly compiled Lean/Rocq proof of the whole statement or a literal object-language ZFC derivation. It proves neither side of the consistency question.

1.2 The delivered polynomials

Every polynomial is specified by a finite list of integer-coefficient residuals and the instruction to square and sum that list. All named expressions are polynomial abbreviations, not additional quantified variables. The JSON convention is

$$[c, [i_1, \ldots, i_d]] \quad\longleftrightarrow\quad c\, x_{i_1} \cdots x_{i_d}; \qquad [c, []] \longleftrightarrow c.$$

Repeated indices are fixed nonnegative exponents. No exponentiation primitive, factorial, binomial primitive, free coefficient parameter, congruence side condition, or implicit sign restriction survives in a final equation.

Polynomial	Variables	Degree	Residuals	Monomials	Height
P_{229}, balanced	229	4	176	4,477	7,968
P_{228}, compact	228	4	175	12,001	7,968
P_{207}, split pooling	207	16	146	94,294	13,607,065,661
P_{206}, full pooling	206	24	145	not fully expanded	not claimed

"Height" means the maximum absolute coefficient of the fully collected single polynomial. For P_{206} the complete residual list is supplied: 144 quadratic residuals and one degree-12 residual. The latter has 25,144 monomials. Its square need not be expanded to specify the polynomial completely. We deliberately do not report a collected coefficient height for that unperformed expansion.

P_{207} has the same 144 quadratic residuals, a degree-8 residual and a degree-5 residual. Its full expansion is included. The maximum absolute coefficient within the residual lists of both pooled equations is 95,232; this is a different statistic from final polynomial height.

1.3 What the improvements measure

The comparison snapshot is `02a78a08603cbae248c19c23a75907c6c6bd6443`. It was the observed `main` head when the source review began. We inspected the three requested repository locations, the active research indexes, and the later delivered packages; the collection's introductory frontier table lags several of its newest files. In particular, package 173 already supplies a 215-variable quartic, and 213/214-variable quartics with much larger coefficients. We do not compare against obsolete 225-variable records as though those were the whole current frontier.

Matched quartic	Variables	Monomials	Height	Residuals
Package 166 balanced parent	230	4,482	7,968	176
New balanced center	229	4,477	7,968	176
Package 166 compact parent	229	12,006	7,968	175
New compact center	228	12,001	7,968	175

Each center reduction saves exactly one integer coordinate and five collected monomials, with degree, height, and residual count unchanged. The maximum residual coefficient also remains 64. These are strict improvements on the two actual parent equations, not a new record for quartic variable count.

Pooling starts from package 173's 221-coordinate, 160-row source chart, the source of its 215-variable balanced quartic. Replacing ten private certificate coordinates by one or two saves nine or eight coordinates; the same six certified projections then give 206 or 207. The degree increases, so these points do not dominate the existing quartics. The repository also contains nine-positive and ten-integer constructions, and a 37-integer quartic with a very different coefficient architecture [1, 7]. "Simplest" therefore requires choosing a measure: variables, degree, support, coefficient bits, evaluation cost, proof complexity, or some specified combination.

The most reusable contribution here is the arithmetic behind the reductions. The finite quadratic-ring construction is inherited from package 166, but the order-eight square correction strengthens its two-square conclusion. Package 168 already pointed toward multi-base reciprocal pooling; the oriented, zero-base-tolerant theorem below and its noncircular activation argument complete a concrete version of that direction. Literature-wide novelty is not established.

2 One scaled square for a cofinal Pell center

For $a \geq 2$ and $j \geq 0$ define the standard Pell polynomials by

$$(a + \sqrt{a^2-1})^j = X_j(a) + Y_j(a)\sqrt{a^2-1}.$$

Equivalently, $X_0 = 1$, $Y_0 = 0$, $X_1 = a$, $Y_1 = 1$, and either sequence satisfies $U_{j+2} = 2aU_{j+1} - U_j$. The identities used below follow by polynomial multiplication. The finite quadratic algebra notation is only a proof device; it does not enlarge the final polynomial's domain.

Theorem 2.1 (One scaled square)**.** *Let $W \geq 2$ be even. Let q, M be positive odd integers, with every prime divisor of M dividing q. Write $n = q + 1$, and put*

$$a_0 \equiv \frac{n^2+1}{2n} = 1 + \frac{q^2}{2(q+1)} \pmod{M}.$$

Then there exist an odd positive integer k and an integer u such that

$$L = X_{Wk}(W+1), \qquad A = L^2 + 2u^2, \qquad A \equiv a_0 \pmod{M}. \tag{2}$$

For this fixed k, the center A can be made arbitrarily large. Moreover $Y_{Wk}(W+1)/W$ is an odd positive integer. Thus, for some integer z, the inherited frame equation is preserved:

$$L^2 - W^3(W+2)(2z+1)^2 = 1. \tag{3}$$

Proof. If $M = 1$, choose $k = 1$ and let u grow; the parity assertion follows from the argument at the end. Otherwise put $D = W(W+2)$ and work in

$$S_M = (\mathbb{Z}/M\mathbb{Z})[s]/(s^2 - D).$$

The element $e = (W+1) + s$ has norm one and inverse $(W+1) - s$. Its Wth power has finite order $2^r k$, with k odd. Therefore $g = e^{Wk}$ has power-of-two order. Write $g = L + Ys$. These are precisely the integer Pell coordinates in (2), reduced modulo M.

Fix $p^a \| M$, where p is odd. We shall prove that

$$c = \frac{a_0 - L^2}{2} \pmod{p^a}$$

is a square. Inverses of all displayed denominators exist because $p \mid q$ and $p \nmid 2(q+1)$.

Case 1: $L \equiv \epsilon \pmod p$, with $\epsilon = \pm 1$. If $D \not\equiv 0 \pmod p$, the norm relation $L^2 - DY^2 = 1$ gives $Y = 0 \pmod p$. If $D = 0 \pmod p$, then $s^2 = 0$. Raising $\epsilon + Ys$ to a power of two shows that its nilpotent coefficient must vanish: that coefficient is an invertible multiple of Y over the odd-characteristic field. Thus $g = \epsilon$ in S_p in either case.

The kernel of reduction from $S_{p^a}^\times$ to $S_p^\times$ consists of elements $1 + pv$ and is a finite p-group. The element ϵg belongs to this kernel and has power-of-two order, so it is the identity. Hence $L^2 = 1 \pmod{p^a}$ and

$$c \equiv \frac{q^2}{4(q+1)} \pmod{p^a}.$$

The unit $q+1$ is 1 modulo p. The root 1 of $b^2 - (q+1)$ has derivative 2 modulo p and lifts successively to a unit root b modulo p^a. Thus $c = (q/(2b))^2$ is a square.

Case 2: $L \not\equiv \pm 1 \pmod p$. The ramified case $D = 0 \pmod p$ is impossible by the same nilpotent argument. Also $Y \neq 0 \pmod p$. In a separable quadratic algebra over $\mathbb{F}_p$, the only norm-one elements with square one are 1 and -1: this is immediate in a field, and in the split algebra the two coordinates are mutual inverses. Therefore the image of g has order at least four. Because W is even and k odd, the order of e modulo p is divisible by eight.

Choose a power $h = H + Zs$ of e of exact order eight. The unique nonidentity norm-one involution is -1, so $h^4 = -1$. Hence $h^{-2} = -h^2$, and comparison of scalar parts gives

$$2H^2 = 1, \qquad DZ^2 = H^2 - 1 = -\tfrac{1}{2}, \qquad -\frac{D}{2} = (DZ)^2 \quad \text{in } \mathbb{F}_p.$$

Since $a_0 = 1 \pmod p$,

$$c = \frac{1 - L^2}{2} = -\frac{DY^2}{2} = (DZY)^2 \pmod p.$$

It is nonzero, so its square root lifts to modulo p^a by the invertibility of $2u$. This proves the local claim without quadratic reciprocity or a formula for the order of the norm-one group.

The Chinese remainder theorem combines the roots into $u \pmod M$. Replacing u by $u + jM$ preserves the congruence and makes $L^2 + 2u^2$ arbitrarily large. In all cases $A \geq L^2$.

It remains to prove the frame parity. At $a = W + 1$, the recurrence modulo W gives $Y_j(a) \equiv j \pmod W$, so $W \mid Y_W(a)$. Because W is even, a is odd and every $X_j(a)$ is odd. The odd-index ordinates are odd; the doubling formula $Y_{2j}(a) = 2X_j(a)Y_j(a)$ therefore gives $\nu_2(Y_W(a)) = \nu_2(W)$. The quotient $Y_W(a)/W$ is odd. Finally,

$$Y_{Wk}(a) = Y_W(a)Y_k(X_W(a)),$$

and the second factor is odd for odd k. The positive quotient is $2z + 1$ for an integer z. Substitution in the Pell norm proves (3). □

2.1 Why the hypotheses matter

The theorem does not say that an arbitrary prescribed center gap is a square or that every preselected frame works. The odd multiplier k is chosen before the correction u. It is an existence choice for already present variables L, z, not a new quantified polynomial coordinate.

There are small exact obstructions to tempting stronger statements. For $W = 5, q = 3, M = 3$, we have $D = 35 = -1 \pmod 3$ and $e = s$. For every odd k, e^{5k} has scalar coordinate $L = 0 \pmod 3$, whereas $a_0 = 1 \pmod 3$. But $2u^2$ is never $1 \pmod 3$. Thus the even-frame hypothesis cannot simply be discarded.

For $W = 6, q = 3, M = 27$, every index $6k$ has $L^2 = 1 \pmod{27}$: in the quadratic algebra, $(7+s)^6 = 1 + 12s$ and $(12s)^2 = 0$ modulo 27, so every power of this element has scalar part one. Here $a_0 = 19 \pmod{27}$. The required correction $18 \pmod{27}$ is not a square, but it equals $2 \cdot 3^2$. A single unscaled square would fail. These examples are also executable negative controls.

3 Applying the center theorem to the literal quartics

3.1 The inherited modular interface

Package 166 directly compiles a numerical proof calculus. Its six exact power requests are

$$R = 16^t, \quad N = B^x, \quad q = (BH)^x, \quad D_t = q^t, \quad D_y = q^{y+1}, \quad M_{\mathrm{pow}} = q^x.$$

Eleven other numerator values are supplied by modular Pell evaluations at base $q+1$. They need a common center satisfying

$$2(q+1)A - (q+1)^2 - 1 \equiv 0 \pmod{\Omega}, \qquad \Omega = NM_{\mathrm{pow}}.$$

In completeness, B, q, Ω are positive odd integers and every prime of Ω divides q. Thus the residue is exactly that of Theorem 2.1. The existing two-square theorem supplies $A = L^2 + u^2 + v^2$; our theorem supplies $A = L^2 + 2u^2$ instead. The signed Pell recognition and common-root modular numerator arguments are inherited [2]. Their source proofs are retained in the archive with their scope and provenance.

3.2 Making the frame even by one constant

Write Σ for the sum of the 21 atom squares in the source frame. The atoms, grouped only for readability, are

$$t, x, y+1, R+t, B; \qquad a_0, \ldots, a_8; \qquad N, J, q, D_t, D_y, M_{\mathrm{pow}}, BH.$$

Here $a_0, \ldots, a_8$ denote actual CRT representatives; the notation a_0 in this paragraph is unrelated to the center residue in Theorem 2.1.

The inherited completeness construction already chooses x, J even, R, y even, one odd CRT representative and eight even ones, and the seven quantities $B, N, q, D_t, D_y, M_{\mathrm{pow}}, BH$ odd. Its later CRT adjustments preserve these parities. The two atoms $t, R+t$ have the same parity, whatever the parity of t. Therefore

$$\Sigma \equiv \underbrace{t^2 + (R+t)^2}_{0} + \underbrace{x^2 + J^2}_{0} + \underbrace{(y+1)^2 + 7}_{0} + \underbrace{\sum_{i=0}^{8} a_i^2}_{1} \equiv 1 \pmod 2.$$

Changing the frame from $W = 2 + \Sigma$ to $W = 3 + \Sigma$ thus makes the canonical completeness frame even. All inherited lower bounds are strengthened. On an arbitrary polynomial zero W need not be even: soundness only uses the stronger magnitude bound and $A \geq L^2$. Evenness is used when constructing witnesses, exactly where the parity calculation applies.

The delivered parity checker reconstructs the complete square sum from the literal raw source. In its stored coordinates the frame constant changes from -525 to -526. It independently propagates the addition chain for J modulo two and checks both possible parities of t. Finite checking authenticates these identities; availability of the canonical semantic choices is the inherited completeness theorem.

3.3 Soundness, completeness, and exact algebra

For soundness, the new center row gives $A \geq L^2$ and the new frame implies every previous frame inequality. All retained arithmetic recognition lemmas apply. The modular numerators, exact binomial coefficients, finite rule rows and Horn-flow certificate decode as in the parent, yielding a C222 refutation.

For completeness, start from a finite C222 refutation. Choose the inherited semantic data with the stated parities, and compute the new even $W = 3 + \Sigma$. Apply Theorem 2.1 with modulus $\Omega = Nq^x$. Choose A large enough for all inherited cap and orientation constructions, then fill their Pell coordinates, modular numerators, and remaining arithmetic gates. This constructs a zero of the modified source. The constrained eliminations and shears have recorded integral restoration maps on their zero sets; deleted unused coordinates can be restored arbitrarily.

The reconstruction replays the parent source's 208 monic eliminations, its integral translations, and the balanced or compact final maps. Every pivot is checked to be an equation $v = f$ with integral f independent of v. Each residual remains quadratic. The now-unused private v is deleted, and every surviving coordinate occurs.

There is also a much shorter independent description at the final endpoint. Let F be the parent's final frame residual after the substitution $v = u$ (in fact it is independent of these center coordinates), and identify its private center coordinates $v = u$. Every residual is then unchanged except $F \mapsto F - 1$. Consequently the complete single polynomials satisfy the exact identity

$$\boxed{P_{\text{new}} = P_{\text{parent}}[v := u] - 2F + 1.} \tag{4}$$

This identity checks the emitted coefficients. It does not by itself prove existential equivalence: an arbitrary old center need not have $u = v$. The cofinal theorem supplies the missing completeness argument by choosing the frame and center afresh.

4 Oriented reciprocal pooling at different bases

The next reduction is independent of the center construction. It starts from package 173's unchanged signed Pell source, whose frame already gives $|A| > W^W$ and bounds the squares of every actual base, requested output, and nominal index by W [4]. The Pell polynomials are evaluated at signed integer A using their polynomial recurrences. For signed indices use $X_{-k}(A) = X_k(A)$ and $Y_{-k}(A) = -Y_k(A)$. These are the coefficients of the corresponding positive or negative powers of the unit $A + s$, $s^2 = A^2 - 1$.

Theorem 4.1 (Oriented pooling with inactive zero bases)**.** *Let $W \in \mathbb{Z}$, $W \geq 258$, let $A \in \mathbb{Z}$ satisfy $|A| > W^W$, and let at most five integer bases n_j be given. There are at most 64 requests in total. Each request in block j consists of integers (m_i, k_i) and canonical signed Pell coordinates (X_i, Y_i) at parameter A and index k_i. Assume*

$$n_j^2 < W, \qquad m_i^2 < W, \qquad k_i^2 < W.$$

Define

$$\begin{aligned} T_j &= 2An_j - n_j^2 - 1, & D_i &= X_i - AY_i, \\ S_j &= \sum_{i \in j} \big((m_i - D_i)^2 - Y_i^2\big), & H_j &= \sum_{i \in j} Y_i(m_i - X_i), \\ E_j &= S_j - 2n_j H_j. \end{aligned}$$

All T_j are nonzero. Then

$$\sum_j \frac{n_j E_j}{T_j} \in \mathbb{Z} \tag{5}$$

if and only if every request in every nonzero-base block satisfies $m_i = n_j^{k_i}$, initially interpreted in $\mathbb{Q}$. For $|n_j| \geq 2$ this also forces $k_i \geq 0$. No conclusion is asserted about a zero-base block's outputs; its term in (5) is zero.

Proof. For $n_j = 0$, $T_j = -1$. For $n_j \neq 0$,

$$|T_j| \geq 2|A| - W > W^W, \qquad \operatorname{sgn} \frac{n_j}{T_j} = \operatorname{sgn} A.$$

Let $\delta_i = m_i - X_i + (A - n_j)Y_i$. Direct expansion gives

$$E_j = \sum_{i \in j} \delta_i^2 + T_j \sum_{i \in j} Y_i^2. \tag{6}$$

Thus (5) is equivalent to integrality of $\sum_j n_j \sum_i \delta_i^2 / T_j$.

We record why signed indices cause no gap. In the quadratic integer ring with $s^2 = A^2 - 1$, send s to $n_j - A$ modulo T_j. This is a ring homomorphism because

$$(n_j - A)^2 - (A^2 - 1) = n_j^2 - 2An_j + 1 = -T_j.$$

The unit $A + s$ maps to n_j, whose inverse is $2A - n_j$, since $n_j(2A - n_j) = 1 \pmod{T_j}$. Therefore for every signed integer k_i,

$$X_i + (n_j - A)Y_i \equiv n_j^{k_i} \pmod{T_j}. \tag{7}$$

Here negative powers on the right mean powers of this modular unit.

Put $b = \lfloor \sqrt{W-1} \rfloor$, so $b \geq 16$. Let J be the set of nonzero-base blocks, $s = |J| \leq 5$, and put

$$C = \prod_{j \in J} n_j^b \neq 0.$$

The integer C appears only in the proof. For a request in an active block define

$$d_i = C m_i - \frac{C}{n_j^b} n_j^{b+k_i} \in \mathbb{Z}.$$

The exponents $b, b+k_i$ are nonnegative. Equation (7) implies $C\delta_i \equiv d_i \pmod{T_j}$, and hence $C^2\delta_i^2 - d_i^2$ is a multiple of T_j. Multiplying the integral sum by C^2 now shows that

$$\Lambda = \sum_{j \in J} \frac{n_j \sum_{i \in j} d_i^2}{T_j}$$

is an integer. This step clears each negative-index denominator before any estimate is used.

Every relevant absolute value is at most b, so

$$|d_i| \leq b^{sb+1} + b^{(s+1)b} \leq 2b^{(s+1)b}.$$

Writing $r \leq 64$ for the total number of requests, we obtain

$$\begin{aligned} \sum_{j \in J} |n_j| \sum_{i \in j} d_i^2 &\leq 4r\, b^{2(s+1)b+1} \\ &\leq 4r\, b^{12b+1} < W^{6b+2} \leq W^W. \end{aligned}$$

The strict inequality follows from $4r \leq 256 < W$ and $b^2 < W$. The last inequality uses $6b+2 \leq b^2+1 \leq W$, valid for $b \geq 16$. Since every denominator has absolute value greater than W^W, $|\Lambda| < 1$. All its nonzero summands have sign $\operatorname{sgn} A$. Thus the integer Λ vanishes, and every $d_i = 0$. Dividing by C gives $m_i = n_j^{k_i}$ in $\mathbb{Q}$. If $|n_j| \geq 2$ and $k_i < 0$, the integer equality $m_i n_j^{-k_i} = 1$ is impossible.

Conversely, exact powers imply $T_j \mid \delta_i$ in every active block. Equation (6) makes E_j/T_j integral there. Every inactive block contributes zero, proving (5). □

4.1 Clearing the denominators without adding a primitive

Introduce one unrestricted integer z and define the ordinary polynomial

$$\mathcal{R} = \sum_{j=0}^{4} n_j E_j \prod_{\substack{0 \leq h \leq 4 \\ h \neq j}} T_h - z \prod_{h=0}^{4} T_h. \tag{8}$$

Since every $T_h \neq 0$ under the retained frame, $\mathcal{R} = 0$ is equivalent to (5). The denominator notation has disappeared from the actual equation. Existing multiplication wires $Z_i = AY_i$ allow the completely literal definition

$$E_j = \sum_{i \in j} \left((m_i - D_i)^2 - Y_i^2 - 2n_j Y_i (m_i - D_i - Z_i) \right).$$

Thus E_j has degree at most three and T_j degree two. Formula (8) has degree at most twelve. Its emitted coefficient list has a nonzero degree-12 part, so its square has exact degree 24.

For the two existing fixed-base pools, the old private variables K_j, L_j satisfy

$$K_j - 2H_j - (2A - n_j)L_j = 0, \qquad S_j - n_j K_j + L_j = 0.$$

Adding n_j times the first row to the second gives $E_j - T_j L_j$. The construction verifies this identity directly against the literal parent rows; the block membership is not inferred from descriptive variable names.

4.2 Activating every block without assuming a positive radix

The five actual bases are

$$t, \quad B, \quad BH, \quad q, \quad q+1,$$

with respectively $1, 1, 1, 3, 10$ requests. All their canonical signed Pell coordinates and the frame bounds are recovered before any deleted power row is used. The unchanged source supplies

$$t = 128 + 4h^2, \qquad K = t^4, \qquad F \text{ requested as } t^K,$$
$$C = 2c, \qquad B\mu = C + F + 1, \qquad H = 2 + 4\sum(\text{actual CRT representatives}),$$
$$q \text{ requested as } (BH)^x.$$

The following order is essential.

1. Since $t \geq 128$, its block is active. Theorem 4.1 gives $F = t^K$. In particular F is even.
2. The literal $C = 2c$ is even. Hence $C + F + 1$ is odd and nonzero. The retained equality $B\mu = C + F + 1$ forces B odd and nonzero, without using the sign of C.
3. $H \equiv 2 \pmod 4$, so $H \neq 0$ and $|H| \geq 2$. The active BH block gives $x \geq 0$ and $q = (BH)^x$.
4. If $x = 0$, then $q = 1$. If $x \geq 1$, then q is nonzero and even. Thus $q \notin \{0, -1\}$, and both remaining bases are nonzero.

All five blocks are now active and all original power assertions hold. The proof has not changed a CRT representative, parity condition, radix constructor, or shared residual to obtain this conclusion.

For a new zero, restore each singleton modulus as T_j and its old quotient as $-\delta_j/T_j$. For a fixed-base block restore

$$L_j = E_j/T_j, \qquad K_j = 2H_j + (2A - n_j)L_j.$$

These are integers by the recovered powers. Every retained coordinate is unchanged. Conversely, at an old zero the old singleton quotient Q_j gives $\delta_j = -T_jQ_j$, so $E_j/T_j = T_jQ_j^2 + Y_j^2$. The explicit polynomial choice

$$z = \sum_{j=0}^{2} n_j(T_jQ_j^2 + Y_j^2) + qL_q + (q+1)L_{q+1}$$

satisfies the new certificate. Thus the transformation is pointwise existential equivalence over the shared source coordinates.

4.3 A degree-16 companion

Use oriented pooling only on the first three bases, with quotient z_0:

$$\mathcal{R}_0 = \sum_{j=0}^{2} n_jE_j \prod_{\substack{0\leq h\leq 2\\ h\neq j}} T_h - z_0 \prod_{h=0}^{2} T_h.$$

The preceding parity argument already makes $q, q+1$ nonzero. Two nonzero consecutive integers have the same sign. Their denominators therefore have the same sign, and the unweighted two-base certificate from package 168 applies [3]:

$$\mathcal{R}_1 = E_3T_4 + E_4T_3 - z_1T_3T_4. \tag{9}$$

For completeness from an old zero take $z_0 = \sum_{j=0}^{2} n_j(T_jQ_j^2 + Y_j^2)$ and $z_1 = L_q + L_{q+1}$. For soundness the same bounded-error proof uses $C = (q(q+1))^b$ and the estimate

$$4 \cdot 13\, b^{6b} < W^{3b+1} \leq W^W.$$

There are no opposite-sign errors to cancel. The first new residual has exact degree eight, the second exact degree five. Their squares and the 144 retained quadratic squares give exact degree sixteen.

4.4 Final variable counts and exact projections

In the parent 221-coordinate chart the removed private coordinates are

$$145, 146, 147, 148, 149, 150, 215, 216, 217, 218,$$

and the removed rows are

$$102, 103, 104, 105, 106, 107, 153, 154, 155, 156.$$

The checker verifies that none of these coordinates occurs in a retained row. The one-certificate source has $221 - 10 + 1 = 212$ variables and 151 rows; the split source has 213 variables and 152 rows.

The same five integral projections along directions in the common kernel of the quadratic Hessians, followed by the same affine consequence, remove six syntax coordinates. At every step the actual pivot slope is 1 or -1, the restoration has integer coefficients, and the new pooling expressions are invariant along the projected directions. The independent checker expands the shears directly rather than using the producer's directional-derivative method. Hence the final counts are 206 and 207.

Both equations share exactly 205 coordinates $x_0, \ldots, x_{204}$, the same 144 quadratic residuals and the same sixteen request tuples. For P_{206} the only new coordinate is $z = x_{205}$; for P_{207} they are $z_0 = x_{205}, z_1 = x_{206}$. Appendix B specifies all those residuals and request tuples literally, so its two short formulas define both complete integer polynomials without importing any unspecified universal polynomial.

5 Logical calibration and formalization boundary

The two source families directly encode the C222 numerical calculus; neither construction simulates K215 or instantiates Jones's universal polynomial. The inherited chain is

$$\begin{aligned} \text{integer polynomial zero} &\longleftrightarrow \text{finite square-shell C222 refutation} \\ &\longleftrightarrow \text{original C222 derivation of code 1} \\ &\longleftrightarrow \text{finite ZF refutation} \\ &\longleftrightarrow \text{finite ZFC refutation.} \end{aligned}$$

The square-shell syntax change is an effective recoding, not an assertion that its numerical constructors literally equal the older Lean ones. The two logical interfaces are

```
KEndpoint.c222_iff_zf_refutation
ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

Their inspected source files are included [5, 6]. The numerical-calculus endpoint has no Turing-machine argument, even though its source file also contains machine-specific theorems.

The full C222/ZF calibration is a substantial inherited theorem. The lower direction uses a model construction through a tagged domain and an internally well-founded quotient component; it is not the invalid assertion that syntactic consistency yields a transitive set model. The ZF/ZFC step is the constructible-model equiconsistency argument. The archive identifies these dependencies but does not recompile their entire transitive Lean import closure. Inspecting two endpoint proof terms alone would not establish every imported theorem's assumptions.

On the arithmetic side, the inherited nontrivial inputs include signed Pell classification and canonical-index recovery, the common frame and orientation bound, generalized binomial CRT coding, the local proof-calculus compiler, and large-evaluation multiplicity recovery. The new proofs state their exact interfaces and preserve or strengthen their hypotheses. The pooled replacement is even pointwise existentially equivalent over all retained source coordinates.

All new proof operations are finite arithmetic, induction, polynomial identities, finite groups/rings, elementary lifting, and CRT. They can be carried out internally in ZFC and do not quantify only over externally standard witnesses. In a nonstandard model the same internal lemmas relate its coded polynomial tuples and coded finite derivations. This is the intended meaning of (1).

6 Reproducibility and evidence

6.1 What is completely specified

The archive includes the LaTeX article and compiled PDF; all four polynomial residual lists; three full collected polynomial expansions; the final-coordinate pool blocks and their factored definitions; exact source changes; integral projection certificates; selected inherited proofs and endpoint source files; independent checkers; an evaluator; and checksums. A finite residual list is already a complete polynomial definition, including for P_{206} whose full squared expansion is not materialized.

6.2 Checks and their proper interpretation

The center endpoint checker imports no producer algebra. It compares every residual by variable names, checks (4), and independently collects every coefficient of both quartics. Six deliberate mutations are rejected. A separate checker authenticates the 21 frame atoms and the exact parity calculation. Two independently implemented finite-ring checks cover 59,880 and 2,040 center congruences; the larger run includes 2,351 nontrivial local order-eight cases. Both mathematical negative controls are retained.

The pooling audit reconstructs the five blocks from the parent rows, checks privacy of all ten deleted coordinates, independently replays the five radical-direction projections and the affine consequence, and compares every final residual coefficient. Its unbounded proof was separately scrutinized, including negative indices, negative centers, unit bases, temporarily zero bases, and the order of the parity bootstrap. Complete finite fixtures supplement, rather than replace, those quantified arguments.

The standard-library entry point is

```
python3 -B verify.py
python3 -B -O verify.py
```

It authenticates the release files when a manifest is present, runs the finite checks, and reconstructs the new equations in temporary directories. The included receipts record the actual completed runs. Disabling Python assertions does not remove the substantive checks: explicit exceptions implement failures.

The evaluator accepts a literal integer tuple of the correct length and computes the chosen polynomial. It is not a root search or a decision procedure for consistency. Rebuild commands and relative paths are in `README.md`; no network is needed for the delivered checks.

7 Research implications and next concrete targets

The first result changes the question asked of a free parameter. The center need not accept every nonnegative slack at a fixed frame; the frame itself can be selected from a cofinal family. Removing the odd part of a finite unit order and using the even frame forces a square class that one scaled square can realize. This suggests examining other free Diophantine parameters whose completeness proofs currently use a universal three- or four-square slack. Any further application must prove its own residue-class and parity conditions; Theorem 2.1 does not permit indiscriminate replacement of nonnegativity witnesses.

The second result makes reciprocal certificates usable before all semantic signs are known. The weight n_j aligns denominator signs, and allowing inactive zero bases separates the recognition lemma from the later nonvanishing proof. The latter is particularly cheap in this source because an even power and an even CRT index already make a retained product odd. A useful next optimization would lower the degree of the five-denominator certificate while preserving this sign and activation argument. Eliminating the orienting factors without a new proof would reintroduce cancellation.

Neither reduction settles the variable/degree/height optimum. The 37-variable large-coefficient quartic and the nine/ten-variable high-degree branches remain separate competitors. For integration, the safest comparison is the literal matched-parent table and the exact source chart used for pooling. An end-to-end formalization of the arithmetic compiler, including the two new lemmas, is a meaningful further target; it is not represented here as already complete.

A Complete balanced quartic P_{229}

All $x_0, \ldots, x_{228}$ range independently over $\mathbb{Z}$. The equation is

$$P_{229}(x_0, \ldots, x_{228}) = \sum_{i=0}^{175} R_i^2 = 0.$$

The following is the entire literal residual list. Variable names in the companion TSV file are explanatory metadata and impose no domain restrictions.

$$R_0 = x_3 - x_4 - 2x_4x_{227}$$

$$R_1 = x_0 - x_0x_2 - x_2 + x_7$$

$$R_2 = -1 - x_2x_5 + x_6x_7 - 2x_{227}$$

$$R_3 = -1 + x_1 - x_6x_7$$

$$R_4 = 1 + x_4 + x_{16} - x_{18} - 2x_{18}x_{227}$$

$$R_5 = -1 - x_8^2 - 2x_8x_9 - x_9^2 + x_{19}$$

$$\begin{aligned} R_6 = {} & -10x_8 + x_8^2 - 6x_8x_9 - 8x_8x_{19} - 20x_9 - 11x_9^2 - 16x_9x_{19} \\ & - 15x_{19} - 4x_{19}^2 + x_{20} \end{aligned}$$

$$R_7 = -2 + 2x_8 - x_8^2 - 4x_8x_{20} - 4x_{20} - 4x_{20}^2 + x_{21}$$

$$R_8 = 1 + 6x_8 - 36x_8^2 - 24x_8x_{21} + 2x_{21} - 4x_{21}^2 + x_{22}$$

$$R_9 = -2 - x_9 - x_9^2 - 2x_9x_{10} - x_{10} - x_{10}^2 + x_{23}$$

$$\begin{aligned} R_{10} = {} & -8 + 7x_8 - x_8^2 + 4x_8x_{10} - 4x_8x_{23} - 4x_{10} + 4x_{23} \\ & + x_{24} - x_{46} \end{aligned}$$

$$R_{11} = -1 + x_8 - x_8^2 - 2x_8x_{10} + x_{10} - x_{10}^2 + x_{25}$$

$$\begin{aligned} R_{12} = {} & 23 + 16x_8 - 8x_8x_9 - 8x_8x_{10} - 16x_8x_{19} - 8x_8x_{25} + 19x_9 \\ & + 9x_9^2 - 36x_9x_{10} - 12x_9x_{25} + 16x_{10} + 8x_{10}^2 - 24x_{10}x_{19} - 8x_{19}x_{25} \\ & - x_{20} + x_{26} - x_{31} \end{aligned}$$

$$R_{13} = -4x_{24}^2 - 8x_{24}x_{26} - 4x_{26}^2 + x_{27}$$

$$\begin{aligned} R_{14} = {} & -2 - 8x_8 + 8x_8x_{23} + 8x_8x_{25} - 8x_9 + 8x_9x_{23} + 8x_9x_{25} \\ & - 4x_{19} + 8x_{23} - 4x_{23}^2 - 8x_{23}x_{25} + 8x_{25} - 4x_{25}^2 + x_{28} \end{aligned}$$

$$\begin{aligned} R_{15} = {} & 6 + 4x_8 + 32x_8x_9 + 16x_8x_{19} + 16x_8x_{25} + 8x_8x_{28} + 24x_9 \\ & - 32x_9x_{25} - 16x_9x_{28} + 12x_{19} - 16x_{19}x_{25} - 8x_{19}x_{28} - x_{20} + 8x_{25} \\ & - 16x_{25}^2 - 16x_{25}x_{28} + 4x_{28} - 4x_{28}^2 + x_{29} \end{aligned}$$

$$\begin{aligned} R_{16} = {} & 22 - 5x_8 - x_8^2 - 8x_8x_9 - 12x_8x_{10} - 4x_8x_{23} - 8x_9 \\ & - 16x_9x_{10} - 16x_9x_{23} - 20x_{10} - 20x_{10}^2 - 24x_{10}x_{23} - 28x_{23} - 4x_{23}^2 \\ & + x_{30} \end{aligned}$$

$$\begin{aligned} R_{17} = {} & 19 - 25x_8 - 7x_8^2 - 8x_8x_9 - 30x_8x_{10} - 16x_8x_{25} + 3x_9 \\ & + 5x_9^2 - 4x_9x_{25} - 27x_{10} - 21x_{10}^2 - 24x_{10}x_{25} - 6x_{23} - 17x_{25} \\ & - 4x_{25}^2 + x_{31} \end{aligned}$$

$$R_{18} = 2 - 4x_{30}^2 - 8x_{30}x_{31} - 4x_{31}^2 + x_{32}$$

$$\begin{aligned} R_{19} = {} & -19x_8 + 8x_8^2 + 8x_8x_{19} + 8x_8x_{23} - 46x_9 - 24x_9^2 + 24x_9x_{19} \\ & + 24x_9x_{23} - 2x_{10} + 4x_{19} - 4x_{19}^2 - 8x_{19}x_{23} + 14x_{23} - 4x_{23}^2 \\ & - x_{25} + x_{33} \end{aligned}$$

$$R_{20} = -3 - 6x_8 - 4x_{10} - 2x_{25} - 4x_{33} - 4x_{33}^2 + x_{34}$$

$$
\begin{aligned}
R_{21} &= 8 + 42x_8 + 7x_8^2 + 28x_8x_{20} + 48x_9 - 32x_9^2 - 32x_9x_{19} \\
&\quad + 64x_9x_{20} - 40x_{19}^2 + 32x_{19}x_{20} + 14x_{20} - 4x_{20}^2 + x_{35} + 2x_{38} \\
R_{22} &= 1 + 6x_8 - 36x_8^2 - 24x_8x_{35} + 2x_{35} - 4x_{35}^2 + x_{36} \\
R_{23} &= -x_9^2 - 4x_9x_{36} - 4x_{36}^2 + x_{37} \\
R_{24} &= -6 - 2x_8 - 4x_9 - 14x_{19} - 16x_{19}^2 + x_{38} \\
R_{25} &= 24 + 21x_9 - 21x_9^2 - 18x_9x_{10} - 24x_9x_{23} - 24x_9x_{38} + 3x_{10} \\
&\quad - 9x_{10}^2 - 16x_{10}x_{38} + 3x_{23} - 12x_{23}^2 - 8x_{23}x_{38} + 4x_{38} - 4x_{38}^2 \\
&\quad + x_{39} + 2x_{46} \\
R_{26} &= 16 - 10x_8 - 20x_8^2 - 16x_8x_{10} - 24x_8x_{25} - 24x_8x_{39} - 12x_{10} \\
&\quad - 16x_{10}x_{25} - 16x_{10}x_{39} - 14x_{25} - 4x_{25}^2 - 8x_{25}x_{39} + 4x_{39} - 4x_{39}^2 \\
&\quad + x_{40} \\
R_{27} &= -x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{41} \\
R_{28} &= 2 + x_8^2 - 4x_8x_{41} + 2x_{10} - 2x_{10}^2 - 8x_{10}x_{41} - 2x_{25} \\
&\quad + 4x_{41} - 4x_{41}^2 + x_{42} \\
R_{29} &= -31 - 16x_{10} - 16x_{10}x_{23} + 16x_{10}x_{25} - 8x_{23} + 12x_{23}^2 - 8x_{23}x_{25} \\
&\quad + 24x_{25} - 4x_{25}^2 + x_{43} - 4x_{46} \\
R_{30} &= -4x_{10} - 16x_{25}^2 - 32x_{25}x_{41} - 16x_{25}x_{42} - 16x_{25}x_{43} - 4x_{41} - 16x_{41}^2 \\
&\quad - 16x_{41}x_{42} - 16x_{41}x_{43} - 2x_{42} - 4x_{42}^2 - 8x_{42}x_{43} - 4x_{43}^2 + x_{44} \\
R_{31} &= -16x_8^2 + 8x_8x_{10} + 32x_8x_{25} + 16x_8x_{28} + x_{10} - x_{10}^2 - 8x_{10}x_{25} \\
&\quad - 4x_{10}x_{28} - 16x_{25}^2 - 16x_{25}x_{28} - 4x_{28}^2 + x_{45} \\
R_{32} &= -1 - 8x_{10} - 4x_{10}^2 + 8x_{10}x_{23} + 8x_{23} - 4x_{23}^2 + x_{46} \\
R_{33} &= -1 - 9x_{10}^2 + 24x_{10}x_{23} + 12x_{10}x_{46} - 16x_{23}^2 - 16x_{23}x_{46} - 4x_{46}^2 \\
&\quad + x_{47} \\
R_{34} &= 1 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} \\
&\quad - 4x_{47}^2 + x_{48} \\
R_{35} &= -1 - 4x_{48} - 4x_{48}^2 + x_{49} \\
R_{36} &= -1 - x_{10}^2 - 8x_{10}x_{48} - 4x_{10}x_{49} - 16x_{48}^2 - 16x_{48}x_{49} - 4x_{49}^2 \\
&\quad + x_{50} \\
R_{37} &= 2 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{50} - 4x_{10}^2 - 8x_{10}x_{50} \\
&\quad - 4x_{50}^2 + x_{51} \\
R_{38} &= -4x_{45}^2 - 8x_{45}x_{51} - 4x_{51}^2 + x_{52} \\
R_{39} &= -9 + 7x_8 - x_8^2 - 8x_8x_{45} - 4x_8x_{52} + 24x_{45} - 16x_{45}^2 \\
&\quad - 16x_{45}x_{52} + 12x_{52} - 4x_{52}^2 + x_{53} \\
R_{40} &= -1 - 4x_{53} - 4x_{53}^2 + x_{54} \\
R_{41} &= -x_{10}^2 - 8x_{10}x_{53} - 4x_{10}x_{54} - 16x_{53}^2 - 16x_{53}x_{54} - 4x_{54}^2 + x_{55} \\
R_{42} &= -x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{55} - 4x_{10} - 4x_{10}^2 - 8x_{10}x_{55} \\
&\quad - 4x_{55} - 4x_{55}^2 + x_{56} \\
R_{43} &= -10 - 12x_{56} - 4x_{56}^2 + x_{57} \\
R_{44} &= 2 + 4x_{10} + 16x_{10}x_{56} + 8x_{10}x_{57} - 4x_{23} - 16x_{23}x_{56} - 8x_{23}x_{57} \\
&\quad - x_{46} + 8x_{56} - 16x_{56}^2 - 16x_{56}x_{57} + 4x_{57} - 4x_{57}^2 + x_{58} \\
R_{45} &= x_9 - x_9^2 + 8x_9x_{10} - 8x_9x_{23} - 4x_9x_{58} - 16x_{10}^2 + 32x_{10}x_{23} \\
&\quad + 16x_{10}x_{58} - 16x_{23}^2 - 16x_{23}x_{58} - 4x_{58}^2 + x_{59} \\
R_{46} &= -10 - 12x_{51} - 4x_{51}^2 + x_{60}
\end{aligned}
$$

$$R_{47} = 1 - 16x_{51}^2 - 16x_{51}x_{59} - 16x_{51}x_{60} - 4x_{59}^2 - 8x_{59}x_{60} - 4x_{60}^2 + x_{61}$$

$$R_{48} = x_{10} - x_{10}^2 - 8x_{10}x_{59} - 4x_{10}x_{61} - 16x_{59}^2 - 16x_{59}x_{61} - 4x_{61}^2 + x_{62}$$

$$R_{49} = -6x_8 - 3x_8^2 - 18x_8x_9 - 16x_9 - 27x_9^2 + 4x_9x_{10} + 3x_{10} + 3x_{10}^2 - 4x_{10}x_{19} - 19x_{19} - 8x_{19}^2 + x_{20} + x_{63} - 4x_{76}$$

$$R_{50} = 1 - 8x_8 - 4x_8x_{10} + 8x_8x_{63} - 16x_9 - 8x_9x_{10} + 16x_9x_{63} + x_{10} + x_{10}^2 + 4x_{10}x_{19} + 8x_{19} - 8x_{19}x_{63} + 3x_{63} - 4x_{63}^2 + x_{64}$$

$$R_{51} = -2 - 12x_8 - 16x_8x_{10} + 16x_8x_{62} + 16x_8x_{64} - 24x_9 - 32x_9x_{10} + 32x_9x_{62} + 32x_9x_{64} + 4x_{10} + 4x_{10}^2 + 16x_{10}x_{19} + 12x_{19} - 16x_{19}x_{62} - 16x_{19}x_{64} - 4x_{62}^2 - 8x_{62}x_{64} - 4x_{63} - 2x_{64} - 4x_{64}^2 + x_{65}$$

$$R_{52} = -2 - 4x_{65} - 4x_{65}^2 + x_{66}$$

$$R_{53} = -x_{13} - 2x_{13}x_{227} + x_{67}$$

$$R_{54} = x_{12} - x_{13} - 2x_{22}x_{67} - x_{68}$$

$$R_{55} = x_{12} - x_{13} - 4x_{24}x_{67} - 2x_{27}x_{67} - x_{69}$$

$$R_{56} = -2x_{29}x_{67} + 2x_{44}x_{67} - 2x_{67} - x_{70} + x_{75}$$

$$R_{57} = x_{12} - x_{13} - 4x_{30}x_{67} - 2x_{32}x_{67} - x_{71}$$

$$R_{58} = x_{12} - x_{13} - 2x_{34}x_{67} + 6x_{67} - x_{72}$$

$$R_{59} = x_{12} - x_{13} - 4x_{36}x_{67} - 2x_{37}x_{67} - x_{73}$$

$$R_{60} = x_{12} - x_{13} - 2x_{40}x_{67} - x_{74}$$

$$R_{61} = x_{12} - x_{13} - 2x_{44}x_{67} - 4x_{67} - x_{75}$$

$$R_{62} = -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 + x_{76}$$

$$R_{63} = 1 + 2x_{11} + x_{11}^2 - x_{76}^2 + x_{77}$$

$$R_{64} = -x_{77}^2 + x_{78}$$

$$R_{65} = -4x_{65}x_{67} - 2x_{66}x_{67} + 2x_{67} + x_{79}$$

$$R_{66} = -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{79} - x_{13}^2 - 2x_{13}x_{79} - x_{79}^2 + x_{80}$$

$$R_{67} = -1 - x_9 - 2x_9x_{227} + x_{81}$$

$$R_{68} = -x_{13}x_{81} + x_{83}x_{84} + x_{85}$$

$$R_{69} = -x_8 - 2x_8x_{227} + x_{82} + 2x_{227}$$

$$R_{70} = 1 + 2x_8 + 4x_8x_{227} + 4x_9 + 8x_9x_{227} - 2x_{19} - 4x_{19}x_{227} + x_{83} + 4x_{227}$$

$$R_{71} = -x_{12}x_{82} + x_{84}$$

$$R_{72} = x_{12}x_{81} + 6x_{13} - 8x_{13}x_{81} - 4x_{13}x_{82} - x_{13}x_{83} - 12x_{13}x_{227} + x_{86}$$

$$R_{73} = -x_{12}x_{68} + x_{13}x_{68} + x_{87}$$

$$R_{74} = -x_{69}x_{87} + x_{88}$$

$$R_{75} = -x_{70}x_{88} + x_{89}$$

$$R_{76} = -x_{71}x_{89} + x_{90}$$

$$R_{77} = -x_{72}x_{90} + x_{91}$$

$$R_{78} = -x_{73}x_{91} + x_{92}$$

$$R_{79} = -x_{74}x_{92} + x_{93}$$

$$R_{80} = -x_{75}x_{93} + x_{94}$$

$$R_{81} = -x_{85}x_{94} + x_{95}$$

$$R_{82} = -x_{86}x_{95} + x_{96}$$

$$R_{83} = -x_{96}^2 + x_{97}$$

$$R_{84} = -x_2 - x_2x_{16} + x_{16} + x_{98}$$

$$R_{85} = -x_{14}x_{98} + x_{99}$$

$$\begin{aligned} R_{86} = &-2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{99} + 2x_1x_{100} - x_{12}^2 - 2x_{12}x_{99} \\ &- x_{13}^2 - 2x_{13}x_{100} + x_{17}x_{102} - x_{78}x_{97} - x_{80}x_{97} - x_{99}^2 - x_{100}^2 \end{aligned}$$

$$R_{87} = -x_2x_{15} + x_{15} - x_{15}x_{98} + x_{100}$$

$$\begin{aligned} R_{88} = &-x_8 - x_9 - 4x_{19} - 2x_{22} - 4x_{24} - 2x_{27} - 2x_{29} \\ &- 4x_{30} - 2x_{32} - 2x_{34} - 4x_{36} - 2x_{37} - 2x_{40} - 2x_{44} \\ &- 4x_{65} - 2x_{66} + x_{101} \end{aligned}$$

$$R_{89} = -3x_{105} - 6x_{105}x_{221} + x_{107}$$

$$R_{90} = x_{103}x_{212} - x_{106} - x_{107}x_{128} - x_{128} - 2x_{128}x_{227}$$

$$R_{91} = x_{108}x_{213} - x_{110}x_{131} - x_{111}$$

$$R_{92} = x_{112}x_{213} - x_{113}x_{131} - x_{114}$$

$$R_{93} = x_{115}x_{213} - x_{116}x_{131} - x_{117}$$

$$R_{94} = x_{118}x_{213} - x_{119}x_{131} - x_{120}$$

$$R_{95} = x_{121}x_{214} - x_{123}x_{131} - x_{124}$$

$$R_{96} = x_{125}x_{214} - x_{126}x_{131} - x_{127}$$

$$R_{97} = -x_{17}x_{128} + x_{131}$$

$$R_{98} = -x_{131}x_{135} + x_{134}x_{212} - x_{136}$$

$$R_{99} = -x_{131}x_{138} + x_{137}x_{213} - x_{139}$$

$$R_{100} = x_{140} - 2x_{140}x_{142} + x_{143}$$

$$R_{101} = -x_{140}x_{143} - x_{143} + x_{144}$$

$$R_{102} = -1 + x_{143}^2 - x_{144}^2 + x_{224}^2$$

$$R_{103} = -x_{141}x_{146} - 7x_{146} + x_{147}$$

$$R_{104} = 2x_{145} + x_{145}^2 + x_{146}^2 - x_{147}^2$$

$$R_{105} = 1 - x_4 - 32x_{141}x_{148} + x_{145} + 16x_{146} - x_{147} + 33x_{148}$$

$$R_{106} = -12 - 2x_{141} - 4x_{141}x_{227} + x_{149} - 24x_{227} + 4x_{227}^2$$

$$R_{107} = x_{17} + x_{149}x_{150} - x_{183} - x_{184} - 2x_{184}x_{227} + x_{185}$$

$$R_{108} = -1 + 2x_{141}x_{196} - x_{151} + 14x_{196} - x_{196}^2$$

$$R_{109} = 3 + x_{151}x_{152} - x_{183} - x_{184}x_{196} + x_{185} + 6x_{221}$$

$$R_{110} = -x_{141}x_{154} - 7x_{154} + x_{155}$$

$$R_{111} = 2x_{153} + x_{153}^2 + x_{154}^2 - x_{155}^2$$

$$R_{112} = -39 + x_{131}x_{228} - 8x_{141} - 12x_{141}x_{221} - 36x_{221} + 36x_{221}^2$$

$$R_{113} = 1 - x_{103} - x_{141}x_{154} + x_{153} - 3x_{154} + 6x_{154}x_{221}$$

$$R_{114} = 7 - x_{131}x_{228} + 2x_{141} + x_{156} - 12x_{221}$$

$$R_{115} = -1 + x_{104} - x_{145} - 3x_{146} - 6x_{146}x_{221} + x_{147} + x_{156}x_{157}$$

$$R_{116} = 7 + x_{141} - x_{141}x_{159} - 7x_{159} + x_{160}$$

$$R_{117} = 1 + 2x_{158} + x_{158}^2 - 2x_{159} + x_{159}^2 - x_{160}^2$$

$$R_{118} = 4 - x_{108} + x_{141} - x_{141}x_{159} + x_{158} - 3x_{159} + 6x_{159}x_{221} - 6x_{221}$$

$$R_{119} = -3x_{104} - 6x_{104}x_{221} + x_{109}$$

$$R_{120} = -x_{141}x_{162} - 7x_{162} + x_{163}$$

$$R_{121} = 2x_{161} + x_{161}^2 + x_{162}^2 - x_{163}^2$$

$$R_{122} = 1 - x_{112} - x_{141}x_{162} + x_{161} - 3x_{162} + 6x_{162}x_{221}$$

$$R_{123} = -x_{141}x_{165} - 7x_{165} + x_{166}$$

$$R_{124} = 2x_{164} + x_{164}^2 + x_{165}^2 - x_{166}^2$$

$$R_{125} = 1 - x_{115} - x_{141}x_{165} + x_{164} - 3x_{165} + 6x_{165}x_{221}$$

$$R_{126} = -x_{141}x_{168} - 7x_{168} + x_{169}$$

$$R_{127} = 2x_{167} + x_{167}^2 + x_{168}^2 - x_{169}^2$$

$$R_{128} = 1 - x_{118} - x_{141}x_{168} + x_{167} - 3x_{168} + 6x_{168}x_{221}$$

$$R_{129} = -x_{141}x_{171} - 7x_{171} + x_{172}$$

$$R_{130} = 2x_{170} + x_{170}^2 + x_{171}^2 - x_{172}^2$$

$$R_{131} = 1 - x_{121} - x_{141}x_{171} + x_{170} - 3x_{171} + 6x_{171}x_{221}$$

$$R_{132} = -x_{141}x_{174} - 7x_{174} + x_{175}$$

$$R_{133} = 2x_{173} + x_{173}^2 + x_{174}^2 - x_{175}^2$$

$$R_{134} = -1 + x_{122} + x_{156}x_{176} - x_{173} - 3x_{174} - 6x_{174}x_{221} + x_{175}$$

$$R_{135} = -x_{141}x_{178} - 7x_{178} + x_{179}$$

$$R_{136} = 2x_{177} + x_{177}^2 + x_{178}^2 - x_{179}^2$$

$$R_{137} = 1 - x_{125} - x_{141}x_{178} + x_{177} - 3x_{178} + 6x_{178}x_{221}$$

$$R_{138} = -x_{141}x_{181} - 7x_{181} + x_{182}$$

$$R_{139} = 2x_{180} + x_{180}^2 + x_{181}^2 - x_{182}^2$$

$$R_{140} = 1 - x_{129}x_{131} - x_{130} - x_{141}x_{181} + x_{180} - 3x_{181} + 6x_{181}x_{221}$$

$$R_{141} = 1 - x_{141}x_{184} - 7x_{184} + x_{185}$$

$$R_{142} = -1 + 2x_{183} + x_{183}^2 + x_{184}^2 - 2x_{185} - x_{185}^2$$

$$R_{143} = x_{128} + x_{156}x_{186} - x_{183} - 3x_{184} - 6x_{184}x_{221} + x_{185}$$

$$R_{144} = -x_{141}x_{188} - 7x_{188} + x_{189}$$

$$R_{145} = 2x_{187} + x_{187}^2 + x_{188}^2 - x_{189}^2$$

$$R_{146} = 1 - x_{131}x_{132} - x_{133} - x_{141}x_{188} + x_{187} - 3x_{188} + 6x_{188}x_{221}$$

$$R_{147} = -x_{141}x_{191} - 7x_{191} + x_{192}$$

$$R_{148} = 2x_{190} + x_{190}^2 + x_{191}^2 - x_{192}^2$$

$$R_{149} = 1 - x_{134} - x_{141}x_{191} + x_{190} - 3x_{191} + 6x_{191}x_{221}$$

$$R_{150} = -x_{141}x_{194} - 7x_{194} + x_{195}$$

$$R_{151} = 2x_{193} + x_{193}^2 + x_{194}^2 - x_{195}^2$$

$$R_{152} = 1 - x_{137} - x_{141}x_{194} + x_{193} - 3x_{194} + 6x_{194}x_{221}$$

$$\begin{aligned} R_{153} = & -13 - 2x_0^2 - 2x_0x_4 - x_1^2 - x_2^2 - x_4^2 + 2x_8 \\ & + 2x_8x_9 - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 \\ & - x_{16}^2 - x_{17}^2 - x_{19} - x_{101}^2 - x_{104}^2 - x_{122}^2 - x_{128}^2 \\ & + x_{140} - x_{196}^2 - 36x_{221} - 36x_{221}^2 - 4x_{227} - 4x_{227}^2 \end{aligned}$$

$$R_{154} = x_{196} - x_{226} - 2x_{226}x_{227}$$

$$R_{155} = -16 + x_0 - x_{197}^2$$

$$R_{156} = -x_{141}x_{199} - 7x_{199} + x_{208}$$

$$R_{157} = -1 + x_{198}^2 + x_{199}^2 - x_{208}^2$$

$$R_{158} = -x_{201} - x_{201}x_{202} + x_{209}$$

$$R_{159} = -1 + x_{200}^2 + x_{201}^2 - x_{209}^2$$

$$R_{160} = -x_{194}^2 + x_{210}$$

$$R_{161} = x_{199} - 2x_{204}x_{210} - x_{210}$$

$$R_{162} = -4x_{199} - 8x_{199}x_{203} + x_{202}$$

$$R_{163} = -6 - x_{141} - x_{198}x_{205} + x_{202}$$

$$R_{164} = -1 - x_{193} - x_{198}x_{206} + x_{200}$$

$$R_{165} = -x_{101} - 4x_{194}x_{207} + x_{201}$$

$$\begin{aligned} R_{166} = & -2 + 2x_0^2 + 2x_0x_4 - 2x_0x_{146} - 2x_0x_{154} + x_1^2 - 2x_1x_{184} \\ & + x_2^2 - 2x_2x_{174} + x_4^2 - 2x_4x_{154} + x_8^2 - 2x_8x_{159} + x_9^2 \\ & - 2x_9x_{162} + x_{10}^2 - 2x_{10}x_{165} + x_{11}^2 - 2x_{11}x_{168} + x_{12}^2 - 2x_{12}x_{171} \\ & + x_{13}^2 - 2x_{13}x_{178} + x_{14}^2 - 2x_{14}x_{181} + x_{15}^2 - 2x_{15}x_{188} + x_{16}^2 \\ & - 2x_{16}x_{191} - 2x_{101}x_{194} - x_{141}x_{211} + x_{145}^2 + x_{153}^2 + x_{158}^2 + 2x_{159} \\ & + x_{161}^2 + x_{164}^2 + x_{167}^2 + x_{170}^2 + x_{173}^2 + x_{177}^2 + x_{180}^2 \\ & + x_{183}^2 + x_{187}^2 + x_{190}^2 + x_{193}^2 + 2x_{194}^2 - 6x_{211} - x_{215}^2 \\ & - x_{216}^2 - x_{217} - x_{217}^2 \end{aligned}$$

$$R_{167} = -x_{104}x_{212} + x_{128}$$

$$R_{168} = -x_{109}x_{213} + x_{128}$$

$$R_{169} = -x_{122}x_{214} + x_{128}$$

$$R_{170} = -1 + x_2 - x_3x_{222}$$

$$\begin{aligned} R_{171} = & -2 - x_{101}^2 - x_{146}^2 - x_{154}^2 + 2x_{159} - x_{159}^2 - x_{162}^2 \\ & - x_{165}^2 - x_{168}^2 - x_{171}^2 - x_{174}^2 - x_{178}^2 - x_{181}^2 - x_{184}^2 \\ & - x_{188}^2 - x_{191}^2 + x_{194}^2 - x_{215}^2 - x_{216}^2 - x_{217} - x_{217}^2 \end{aligned}$$

$$\begin{aligned} R_{172} = & -1 - 4x_{106}^2 - 4x_{111}^2 - 4x_{114}^2 - 4x_{117}^2 - 4x_{120}^2 - 4x_{124}^2 \\ & - 4x_{127}^2 + x_{128}^2 - 4x_{130}^2 - 4x_{133}^2 - 4x_{136}^2 - 4x_{139}^2 - x_{218}^2 \\ & - x_{219}^2 - x_{220} - x_{220}^2 \end{aligned}$$

$$R_{173} = -1 + x_{222} - 4x_{223} - 4x_{223}^2$$

$$R_{174} = 7 + x_{141} - x_{224}^2 - 2x_{225}^2$$

$$\begin{aligned} R_{175} = & -3 - 2x_0 - 2x_4 + 2x_8 - x_8^2 - x_9^2 - x_{10}^2 \\ & - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 - x_{15}^2 - x_{16}^2 + x_{226} \end{aligned}$$

B Complete definitions of the pooled equations

All coordinates range independently over the integers. These two polynomials share the following 205 coordinates and 144 quadratic residuals. Set

$$A = x_{132} + 1, \qquad (n_0, n_1, n_2, n_3, n_4) = (x_0, x_3, x_{178}, x_{95} - 1, x_{95}).$$

For each row of the following table, use the displayed affine polynomials as (m, D, Y, Z) in its block. The table is part of the polynomial definition; Pell notation is unnecessary to evaluate it.

Block j	Request	m	D	Y	Z
0	0	x_2	x_{175}	x_{176}	x_{177}
1	0	x_{17}	x_{166}	x_{167}	$1 + x_{168}$
2	0	$-1 + x_{95}$	x_{166}	x_{167}	$1 + x_{168}$
3	0	x_{97}	x_{136}	x_{137}	x_{138}
3	1	x_{114}	x_{157}	x_{158}	x_{159}
3	2	x_{121}	x_{166}	x_{167}	$1 + x_{168}$
4	0	x_{96}	x_{139}	x_{140}	x_{141}
4	1	x_{101}	x_{142}	$-1 + x_{143}$	x_{144}
4	2	x_{104}	x_{145}	x_{146}	x_{147}
4	3	x_{107}	x_{148}	x_{149}	x_{150}
4	4	x_{110}	x_{151}	x_{152}	x_{153}
4	5	x_{113}	x_{154}	x_{155}	x_{156}
4	6	x_{117}	x_{160}	x_{161}	x_{162}
4	7	x_{120}	x_{163}	x_{164}	x_{165}
4	8	x_{125}	x_{169}	x_{170}	x_{171}
4	9	x_{128}	x_{172}	x_{173}	x_{174}

Define the following fixed polynomials by finite sums over the table:

$$T_j = 2An_j - n_j^2 - 1, \qquad E_j = \sum_{i \in j} \left((m_i - D_i)^2 - Y_i^2 - 2n_j Y_i (m_i - D_i - Z_i) \right).$$

No table entry is an extra unknown. With $Q_0, \ldots, Q_{143}$ given in full below, the 206-variable equation is

$$P_{206}(x_0, \ldots, x_{205}) = \sum_{i=0}^{143} Q_i^2 + \mathcal{R}^2 = 0,$$
$$\mathcal{R} = \sum_{j=0}^{4} n_j E_j \prod_{\substack{0 \le h \le 4 \\ h \ne j}} T_h - x_{205} \prod_{h=0}^{4} T_h.$$

The 207-variable equation is

$$P_{207}(x_0, \ldots, x_{206}) = \sum_{i=0}^{143} Q_i^2 + \mathcal{R}_0^2 + \mathcal{R}_1^2 = 0,$$
$$\mathcal{R}_0 = \sum_{j=0}^{2} n_j E_j \prod_{\substack{0 \le h \le 2 \\ h \ne j}} T_h - x_{205} T_0 T_1 T_2,$$
$$\mathcal{R}_1 = E_3 T_4 + E_4 T_3 - x_{206} T_3 T_4.$$

These finite formulas, the complete table, and the following list are a standalone specification of both ordinary integer polynomials. In particular the symbolic products have the fixed lengths three or five; there is no variable-length product in a final equation.

$$Q_0 = x_0 - x_0 x_4 - x_4 + x_7$$

$$Q_1 = -1 + x_1 - x_2 - x_4x_5$$

$$Q_2 = -1 + x_1 - x_6x_7$$

$$Q_3 = 1 + x_2 - x_3x_{18} + 2x_{16}$$

$$Q_4 = -1 + 2x_8 - x_8^2 - 2x_8x_9 + 2x_9 - x_9^2 + x_{19}$$

$$Q_5 = -2 + 2x_8 - x_8^2 - 4x_8x_{20} + 4x_{20} - 4x_{20}^2 + x_{21}$$

$$\begin{aligned} Q_6 = 1 &+ 6x_8 - 36x_8^2 - 192x_8x_{20} - 24x_8x_{21} + 16x_{20} - 256x_{20}^2 \\ &- 64x_{20}x_{21} + 2x_{21} - 4x_{21}^2 + x_{22} \end{aligned}$$

$$Q_7 = -2 - x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{23}$$

$$Q_8 = -1 + 2x_8 - x_8^2 - 2x_8x_{10} + 2x_{10} - x_{10}^2 + x_{24}$$

$$\begin{aligned} Q_9 = -9 &+ x_8 - 145x_8^2 - 196x_8x_9 - 192x_8x_{10} - 48x_8x_{19} - 4x_8x_{23} \\ &- 48x_8x_{24} + 4x_9 - 68x_9^2 - 128x_9x_{10} - 32x_9x_{19} - 8x_9x_{23} - 32x_9x_{24} \\ &- 64x_{10}^2 - 32x_{10}x_{19} - 32x_{10}x_{24} - 2x_{19} - 4x_{19}^2 - 8x_{19}x_{24} + 12x_{23} \\ &- 4x_{23}^2 - 4x_{24}^2 + x_{25} \end{aligned}$$

$$\begin{aligned} Q_{10} = -18 &+ 14x_8 - 2x_8^2 - 8x_8x_9 - 8x_8x_{23} + 24x_9 - 8x_9^2 \\ &- 16x_9x_{23} + 24x_{23} - 8x_{23}^2 - 4x_{25}^2 + x_{26} \end{aligned}$$

$$\begin{aligned} Q_{11} = -6 &+ 16x_{10} - 16x_{10}^2 - 16x_{10}x_{23} - 16x_{10}x_{24} + 8x_{23} - 4x_{23}^2 \\ &- 8x_{23}x_{24} + 8x_{24} - 4x_{24}^2 + x_{27} \end{aligned}$$

$$\begin{aligned} Q_{12} = 1 &+ 6x_8 - 36x_8^2 - 96x_8x_9 - 48x_8x_{10} - 24x_8x_{19} - 48x_8x_{24} \\ &- 24x_8x_{27} + 8x_9 - 64x_9^2 - 64x_9x_{10} - 32x_9x_{19} - 64x_9x_{24} - 32x_9x_{27} \\ &+ 8x_{10} - 16x_{10}^2 - 16x_{10}x_{19} - 32x_{10}x_{24} - 16x_{10}x_{27} + 2x_{19} - 4x_{19}^2 \\ &- 16x_{19}x_{24} - 8x_{19}x_{27} + 8x_{24} - 16x_{24}^2 - 16x_{24}x_{27} + 4x_{27} - 4x_{27}^2 \\ &+ x_{28} \end{aligned}$$

$$\begin{aligned} Q_{13} = -12 &- 29x_8 - 37x_8^2 - 24x_8x_9 - 112x_8x_{10} - 4x_8x_{23} - 24x_8x_{24} \\ &- 39x_9 - 37x_9^2 - 112x_9x_{10} - 24x_9x_{23} - 4x_9x_{24} - 80x_{10} - 128x_{10}^2 \\ &- 32x_{10}x_{23} - 32x_{10}x_{24} - 12x_{23} - 4x_{23}^2 - 8x_{24} - 4x_{24}^2 + x_{29} \end{aligned}$$

$$\begin{aligned} Q_{14} = -18 &- 10x_8 - 2x_8^2 - 24x_8x_9 - 32x_8x_{10} - 8x_8x_{23} - 72x_9 \\ &- 72x_9^2 - 192x_9x_{10} - 48x_9x_{23} - 96x_{10} - 128x_{10}^2 - 64x_{10}x_{23} - 24x_{23} \\ &- 8x_{23}^2 - 4x_{29}^2 + x_{30} \end{aligned}$$

$$\begin{aligned} Q_{15} = -12 &+ 16x_8 - 4x_8^2 - 8x_8x_{10} - 8x_8x_{19} - 8x_8x_{23} + 16x_{10} \\ &- 4x_{10}^2 - 8x_{10}x_{19} - 8x_{10}x_{23} + 16x_{19} - 4x_{19}^2 - 8x_{19}x_{23} + 16x_{23} \\ &- 4x_{23}^2 + x_{31} \end{aligned}$$

$$\begin{aligned} Q_{16} = -3 &- 24x_8 - 64x_8^2 - 160x_8x_{10} - 64x_8x_{23} - 32x_8x_{24} - 32x_8x_{31} \\ &- 26x_{10} - 100x_{10}^2 - 80x_{10}x_{23} - 40x_{10}x_{24} - 40x_{10}x_{31} - 8x_{23} - 16x_{23}^2 \\ &- 16x_{23}x_{24} - 16x_{23}x_{31} - 6x_{24} - 4x_{24}^2 - 8x_{24}x_{31} - 4x_{31} - 4x_{31}^2 \\ &+ x_{32} \end{aligned}$$

$$\begin{aligned} Q_{17} = -5 &- 30x_8 - 36x_8^2 - 96x_8x_9 - 24x_8x_{19} - 40x_9 - 64x_9^2 \\ &- 32x_9x_{19} - 10x_{19} - 4x_{19}^2 + x_{20} \end{aligned}$$

$$\begin{aligned} Q_{18} = -82 &- 414x_8 - 529x_8^2 - 1472x_8x_9 - 368x_8x_{19} + 92x_8x_{20} - 576x_9 \\ &- 1024x_9^2 - 512x_9x_{19} + 128x_9x_{20} - 144x_{19} - 64x_{19}^2 + 32x_{19}x_{20} + 36x_{20} \\ &- 4x_{20}^2 + x_{33} \end{aligned}$$

$$
\begin{aligned}
Q_{19} = & -4159 - 23994x_8 - 34596x_8^2 - 95232x_8x_9 - 23808x_8x_{19} + 5952x_8x_{20} + 744x_8x_{33} \\
& - 33024x_9 - 65536x_9^2 - 32768x_9x_{19} + 8192x_9x_{20} + 1024x_9x_{33} - 8256x_{19} - 4096x_{19}^2 \\
& + 2048x_{19}x_{20} + 256x_{19}x_{33} + 2064x_{20} - 256x_{20}^2 - 64x_{20}x_{33} + 258x_{33} - 4x_{33}^2 \\
& + x_{34} \\
Q_{20} = & -x_9^2 - 4x_9x_{34} - 4x_{34}^2 + x_{35} \\
Q_{21} = & -6 - 30x_8 - 64x_8^2 - 128x_8x_9 - 64x_8x_{19} - 32x_9 - 64x_9^2 \\
& - 64x_9x_{19} - 14x_{19} - 16x_{19}^2 + x_{36} \\
Q_{22} = & -4 + 8x_9 - 64x_9^2 - 96x_9x_{10} - 32x_9x_{23} - 32x_9x_{36} + 6x_{10} \\
& - 36x_{10}^2 - 24x_{10}x_{23} - 24x_{10}x_{36} + 2x_{23} - 4x_{23}^2 - 8x_{23}x_{36} + 4x_{36} \\
& - 4x_{36}^2 + x_{37} \\
Q_{23} = & 8x_8 - 64x_8^2 - 96x_8x_{10} - 32x_8x_{24} - 32x_8x_{37} + 6x_{10} - 36x_{10}^2 \\
& - 24x_{10}x_{24} - 24x_{10}x_{37} + 2x_{24} - 4x_{24}^2 - 8x_{24}x_{37} + 4x_{37} - 4x_{37}^2 \\
& + x_{38} \\
Q_{24} = & -x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39} \\
Q_{25} = & -35 + 26x_8 - 5x_8^2 - 8x_8x_9 - 4x_8x_{10} - 8x_8x_{23} - 8x_8x_{24} \\
& - 4x_8x_{39} + 24x_9 - 4x_9^2 - 8x_9x_{23} - 8x_9x_{24} + 4x_{10} - 4x_{10}^2 \\
& - 8x_{10}x_{39} + 24x_{23} - 4x_{23}^2 - 8x_{23}x_{24} + 24x_{24} - 4x_{24}^2 + 4x_{39} \\
& - 4x_{39}^2 + x_{40} \\
Q_{26} = & 4x_8 - 18x_8^2 - 40x_8x_{10} - 32x_8x_{24} - 40x_8x_{39} - 16x_8x_{40} + 4x_{10} \\
& - 24x_{10}^2 - 32x_{10}x_{24} - 48x_{10}x_{39} - 16x_{10}x_{40} - 16x_{24}^2 - 32x_{24}x_{39} - 16x_{24}x_{40} \\
& + 4x_{39} - 24x_{39}^2 - 16x_{39}x_{40} - 4x_{40}^2 + x_{41} \\
Q_{27} = & x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{27} - 16x_{24}^2 - 16x_{24}x_{27} - 4x_{27}^2 \\
& + x_{42} \\
Q_{28} = & -1 + 8x_9 - 4x_9^2 - 8x_9x_{23} + 8x_{23} - 4x_{23}^2 + x_{43} \\
Q_{29} = & -1 - 16x_9^2 - 8x_9x_{10} - 32x_9x_{23} - 16x_9x_{43} - x_{10}^2 - 8x_{10}x_{23} \\
& - 4x_{10}x_{43} - 16x_{23}^2 - 16x_{23}x_{43} - 4x_{43}^2 + x_{44} \\
Q_{30} = & 1 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{44} - 4x_{10}^2 - 8x_{10}x_{44} \\
& - 4x_{44}^2 + x_{45} \\
Q_{31} = & -1 - 4x_{45} - 4x_{45}^2 + x_{46} \\
Q_{32} = & -1 - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} - 16x_{45}^2 - 16x_{45}x_{46} - 4x_{46}^2 \\
& + x_{47} \\
Q_{33} = & 2 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} \\
& - 4x_{47}^2 + x_{48} \\
Q_{34} = & -4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49} \\
Q_{35} = & -9 + 7x_8 - x_8^2 - 8x_8x_{42} - 4x_8x_{49} + 24x_{42} - 16x_{42}^2 \\
& - 16x_{42}x_{49} + 12x_{49} - 4x_{49}^2 + x_{50} \\
Q_{36} = & -1 - 4x_{50} - 4x_{50}^2 + x_{51} \\
Q_{37} = & -x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 + x_{52} \\
Q_{38} = & -x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10} - 4x_{10}^2 - 8x_{10}x_{52} \\
& - 4x_{52} - 4x_{52}^2 + x_{53} \\
Q_{39} = & -10 - 12x_{53} - 4x_{53}^2 + x_{54} \\
Q_{40} = & 1 + 4x_9 - 4x_9^2 - 8x_9x_{23} - 16x_9x_{53} - 8x_9x_{54} + 4x_{23} \\
& - 4x_{23}^2 - 16x_{23}x_{53} - 8x_{23}x_{54} + 8x_{53} - 16x_{53}^2 - 16x_{53}x_{54} + 4x_{54} \\
& - 4x_{54}^2 + x_{55}
\end{aligned}
$$

$$
\begin{aligned}
Q_{41} = & -10 + x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{55} - 16x_{23}^2 - 16x_{23}x_{55} \\
& - 12x_{48} - 4x_{48}^2 - 4x_{55}^2 + x_{56} \\
Q_{42} = & \ 21 + 24x_{48} - 8x_{48}^2 - 16x_{48}x_{56} - 2x_{56} - 4x_{56}^2 + x_{57} \\
Q_{43} = & \ 1 + 4x_8 - 4x_8^2 - 8x_8x_{19} - 8x_8x_{59} + x_{10} - x_{10}^2 \\
& - 4x_{10}x_{57} + 4x_{19} - 4x_{19}^2 - 8x_{19}x_{59} - 4x_{57}^2 + x_{58} + 4x_{59} \\
& - 4x_{59}^2 \\
Q_{44} = & -4x_8 - 4x_8^2 - 4x_8x_{10} - 8x_8x_{19} - x_{10} - x_{10}^2 - 4x_{10}x_{19} \\
& - 4x_{19} - 4x_{19}^2 + x_{59} \\
Q_{45} = & \ 4x_8 - 24x_8^2 - 48x_8x_{19} - 16x_8x_{58} - 16x_8x_{59} + 4x_{19} - 24x_{19}^2 \\
& - 16x_{19}x_{58} - 16x_{19}x_{59} - 4x_{58}^2 + 8x_{59} - 8x_{59}^2 + x_{60} \\
Q_{46} = & -2 - 4x_{60} - 4x_{60}^2 + x_{61} \\
Q_{47} = & -x_2x_{13} + x_{62} \\
Q_{48} = & \ x_{12} - x_{13} - 2x_{22}x_{62} - x_{63} \\
Q_{49} = & \ x_{12} - x_{13} - 2x_{26}x_{62} - x_{64} \\
Q_{50} = & \ x_{12} - x_{13} - 2x_{28}x_{62} - 6x_{62} - x_{65} \\
Q_{51} = & \ x_{12} - x_{13} - 2x_{30}x_{62} - x_{66} \\
Q_{52} = & \ x_{12} - x_{13} - 2x_{32}x_{62} + 6x_{62} - x_{67} \\
Q_{53} = & \ x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68} \\
Q_{54} = & \ x_{12} - x_{13} - 2x_{38}x_{62} - x_{69} \\
Q_{55} = & \ x_{12} - x_{13} - 2x_{41}x_{62} - 4x_{62} - x_{70} \\
Q_{56} = & -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 + x_{71} \\
Q_{57} = & \ 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72} \\
Q_{58} = & -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 + x_{73} \\
& - x_{74}^2 \\
Q_{59} = & -4x_{60}x_{62} - 2x_{61}x_{62} + 2x_{62} + x_{74} \\
Q_{60} = & -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77} \\
Q_{61} = & -1 + 2x_2 - 2x_2x_8 - 2x_2x_{19} + x_{75} \\
Q_{62} = & -x_8x_{202} - x_{12} + x_{76} + x_{202} \\
Q_{63} = & -6x_8x_{62} - 8x_9x_{62} + x_9x_{202} + x_{12} - x_{13} - 2x_{19}x_{62} + x_{78} \\
Q_{64} = & -x_{12}x_{63} + x_{13}x_{63} + x_{79} \\
Q_{65} = & -x_{64}x_{79} + x_{80} \\
Q_{66} = & -x_{65}x_{80} + x_{81} \\
Q_{67} = & -x_{66}x_{81} + x_{82} \\
Q_{68} = & -x_{67}x_{82} + x_{83} \\
Q_{69} = & -x_{68}x_{83} + x_{84} \\
Q_{70} = & -x_{69}x_{84} + x_{85} \\
Q_{71} = & -x_{70}x_{85} + x_{86} \\
Q_{72} = & -x_{77}x_{86} + x_{87} \\
Q_{73} = & -x_{78}x_{87} + x_{88} \\
Q_{74} = & -x_{88}^2 + x_{89}
\end{aligned}
$$

$$Q_{75} = -x_4 - 2x_4x_{16} + 2x_{16} + x_{90}$$

$$Q_{76} = -x_{14}x_{90} + x_{91}$$

$$Q_{77} = -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{94} - x_{73}x_{89} - x_{91}^2 - x_{92}^2$$

$$Q_{78} = -x_4x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$

$$Q_{79} = -x_{95}x_{98} + x_{98} + x_{100}$$

$$Q_{80} = -x_3x_{121} + x_{96}x_{194} - x_{99} - x_{100}x_{121}$$

$$Q_{81} = x_{101}x_{194} - x_{102}x_{124} - x_{103}$$

$$Q_{82} = x_{104}x_{194} - x_{105}x_{124} - x_{106}$$

$$Q_{83} = x_{107}x_{194} - x_{108}x_{124} - x_{109}$$

$$Q_{84} = x_{110}x_{194} - x_{111}x_{124} - x_{112}$$

$$Q_{85} = x_{113}x_{195} - x_{115}x_{124} - x_{116}$$

$$Q_{86} = x_{117}x_{195} - x_{118}x_{124} - x_{119}$$

$$Q_{87} = -x_{17}x_{121} + x_{124}$$

$$Q_{88} = x_{120} - x_{122}x_{124} - x_{123}$$

$$Q_{89} = -x_{124}x_{126} + x_{125} - x_{127}$$

$$Q_{90} = -x_{124}x_{129} + x_{128}x_{194} - x_{130}$$

$$Q_{91} = x_{131} - 2x_{131}x_{133} + x_{134}$$

$$Q_{92} = -x_{131}x_{134} - x_{134} + x_{135}$$

$$Q_{93} = 2x_{132} + x_{132}^2 + x_{134}^2 - x_{135}^2$$

$$Q_{94} = -x_{132}x_{137} - x_{137} + x_{138}$$

$$Q_{95} = -1 + x_{136}^2 + 2x_{136}x_{138} + x_{137}^2$$

$$Q_{96} = -x_{132}x_{140} - x_{140} + x_{141}$$

$$Q_{97} = -1 + x_{139}^2 + 2x_{139}x_{141} + x_{140}^2$$

$$Q_{98} = 1 + x_{132} - x_{132}x_{143} - x_{143} + x_{144}$$

$$Q_{99} = x_{142}^2 + 2x_{142}x_{144} - 2x_{143} + x_{143}^2$$

$$Q_{100} = -x_{132}x_{146} - x_{146} + x_{147}$$

$$Q_{101} = -1 + x_{145}^2 + 2x_{145}x_{147} + x_{146}^2$$

$$Q_{102} = -x_{132}x_{149} - x_{149} + x_{150}$$

$$Q_{103} = -1 + x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$Q_{104} = -x_{132}x_{152} - x_{152} + x_{153}$$

$$Q_{105} = -1 + x_{151}^2 + 2x_{151}x_{153} + x_{152}^2$$

$$Q_{106} = -x_{132}x_{155} - x_{155} + x_{156}$$

$$Q_{107} = -1 + x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$

$$Q_{108} = -x_{132}x_{158} - x_{158} + x_{159}$$

$$Q_{109} = -1 + x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$

$$Q_{110} = -x_{132}x_{161} - x_{161} + x_{162}$$

$$Q_{111} = -1 + x_{160}^2 + 2x_{160}x_{162} + x_{161}^2$$

$$Q_{112} = -x_{132}x_{164} - x_{164} + x_{165}$$

$$Q_{113} = -1 + x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$

$$Q_{114} = 1 - x_{132}x_{167} - x_{167} + x_{168}$$

$$Q_{115} = -1 + 2x_{166} + x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$

$$Q_{116} = -x_{132}x_{170} - x_{170} + x_{171}$$

$$Q_{117} = -1 + x_{169}^2 + 2x_{169}x_{171} + x_{170}^2$$

$$Q_{118} = -x_{132}x_{173} - x_{173} + x_{174}$$

$$Q_{119} = -1 + x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$

$$Q_{120} = -x_{132}x_{176} - x_{176} + x_{177}$$

$$Q_{121} = -1 + x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$

$$\begin{aligned} Q_{122} = &-5 - 2x_0 - 4x_0^2 - 2x_0x_2 - x_1^2 - 2x_2^2 - x_3^2 \\ &- x_4^2 + 2x_8 - x_8^2 - x_9^2 - x_{10}^2 - x_{11}^2 - x_{12}^2 \\ &- x_{13}^2 - x_{14}^2 - x_{15}^2 - 4x_{16}^2 - x_{17}^2 - x_{93}^2 + 2x_{95} \\ &- 2x_{95}^2 - x_{96}^2 - x_{97}^2 - x_{101}^2 - x_{104}^2 - x_{107}^2 - x_{110}^2 \\ &- x_{113}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 \\ &+ x_{131} - x_{178}^2 \end{aligned}$$

$$\begin{aligned} Q_{123} = &\, 2x_3 - 4x_3x_8 - 4x_3x_9 - 4x_3x_{10} - 4x_3x_{11} - 4x_3x_{12} - 4x_3x_{13} \\ &- 4x_3x_{14} - 4x_3x_{15} - 8x_3x_{16} + x_{178} \end{aligned}$$

$$Q_{124} = -128 + x_0 - 4x_{179}^2$$

$$Q_{125} = -x_{132}x_{181} - x_{181} + x_{190}$$

$$Q_{126} = -1 + x_{180}^2 + x_{181}^2 - x_{190}^2$$

$$Q_{127} = -x_{183} - x_{183}x_{184} + x_{191}$$

$$Q_{128} = -1 + x_{182}^2 + x_{183}^2 - x_{191}^2$$

$$Q_{129} = -x_{173}^2 + x_{192}$$

$$Q_{130} = x_{181} - 2x_{186}x_{192} - x_{192}$$

$$Q_{131} = -4x_{181} - 8x_{181}x_{185} + x_{184}$$

$$Q_{132} = -x_{132} - x_{180}x_{187} + x_{184}$$

$$Q_{133} = -x_{172} - x_{174} - x_{180}x_{188} + x_{182}$$

$$Q_{134} = -2x_{16} - 4x_{173}x_{189} + x_{183}$$

$$\begin{aligned} Q_{135} = &\, 26 + 2x_0^2 + 2x_0x_2 - 2x_0x_{137} - 2x_0x_{140} + x_1^2 - 2x_1x_{167} \\ &+ x_2^2 - 2x_2x_{140} + x_4^2 - 2x_4x_{158} + x_8^2 - 2x_8x_{143} + x_9^2 \\ &- 2x_9x_{146} + x_{10}^2 - 2x_{10}x_{149} + x_{11}^2 - 2x_{11}x_{152} + x_{12}^2 - 2x_{12}x_{155} \\ &+ x_{13}^2 - 2x_{13}x_{161} + x_{14}^2 - 2x_{14}x_{164} + x_{15}^2 - 2x_{15}x_{170} + 4x_{16}^2 \\ &- 4x_{16}x_{173} + x_{93}^2 - 2x_{93}x_{176} - x_{132}x_{193} - 2x_{136} - 2x_{138} + x_{138}^2 \\ &- 2x_{139} - 2x_{141} + x_{141}^2 - 2x_{142} + 2x_{143} - 2x_{144} + x_{144}^2 \\ &- 2x_{145} - 2x_{147} + x_{147}^2 - 2x_{148} - 2x_{150} + x_{150}^2 - 2x_{151} \\ &- 2x_{153} + x_{153}^2 - 2x_{154} - 2x_{156} + x_{156}^2 - 2x_{157} - 2x_{159} \\ &+ x_{159}^2 - 2x_{160} - 2x_{162} + x_{162}^2 - 2x_{163} - 2x_{165} + x_{165}^2 \\ &- 2x_{166} + x_{168}^2 - 2x_{169} - 2x_{171} + x_{171}^2 - 2x_{172} - 2x_{174} \\ &+ x_{174}^2 - 2x_{175} - 2x_{177} + x_{177}^2 \end{aligned}$$

$$Q_{136} = -x_{97}x_{194} + x_{121}$$

$$
\begin{aligned}
Q_{137} &= -x_{114}x_{195} + x_{121} \\
Q_{138} &= -2 - 4x_{16}^2 - x_{137}^2 - x_{140}^2 + 2x_{143} - x_{143}^2 - x_{146}^2 \\
&\quad - x_{149}^2 - x_{152}^2 - x_{155}^2 - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 \\
&\quad - x_{170}^2 + x_{173}^2 - x_{176}^2 - x_{196}^2 - x_{197}^2 - x_{198} - x_{198}^2 \\
Q_{139} &= -1 - 4x_{99}^2 - 4x_{103}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 - 4x_{116}^2 \\
&\quad - 4x_{119}^2 + x_{121}^2 - 4x_{123}^2 - 4x_{127}^2 - 4x_{130}^2 - x_{199}^2 - x_{200}^2 \\
&\quad - x_{201} - x_{201}^2 \\
Q_{140} &= -x_2x_{12} + x_{202} \\
Q_{141} &= -1 - x_2 - 2x_2x_{203} + x_4 \\
Q_{142} &= -x_0^2 + x_{204} \\
Q_{143} &= x_{93} - x_{204}^2
\end{aligned}
$$

C Companion quartic and machine-readable identities

P_{228} is the sum of the 175 residual squares in `equations/P228_compact/equation.json`; every collected coefficient is also supplied in that directory. Its readable residual list and LaTeX include are complete. It is obtained from the compact package-166 parent by exactly the same two-row change and private variable identification proved above. The balanced and compact parents use different integral projections; one must not delete a coordinate from the balanced list and assume that the compact equation follows without its separate map.

The source-coordinate and final-coordinate certificates define every map, including all restored eliminated variables. The sorted sparse integer lists are authoritative mathematical data; PDF metadata and font bytes may differ across TeX installations without changing any equation. The final checksum manifest covers the supplied files except itself. Historical source notices are preserved, and the new work uses the repository's MIT No Attribution terms.

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